

Broadcom Inc.

Recommendation **BUY** ★★★★★

Price USD 838.36 [as of market close Oct 27, 2023] **12-Mo. Target Price** USD 975.00 **Report Currency** USD **Investment Style** Large-Cap Blend

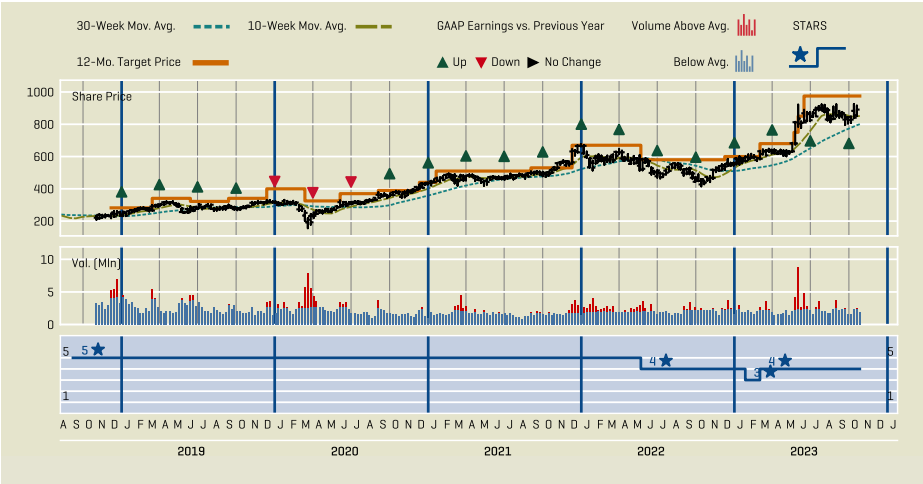
Equity Analyst Angelo Zino, CFA

GICS Sector Information Technology **Sub-Industry** Semiconductors **Summary** Broadcom is a well-diversified designer, developer, and global supplier of a broad range of analog semiconductor devices.

Key Stock Statistics (Source: CFRA, S&P Global Market Intelligence [SPGMI], Company Reports)

52-Wk Range	USD 925.91 - 441.36	Oper.EPS2023E	USD 42.21	Market Capitalization[B]	USD 346.02	Beta	1.14
Trailing 12-Month EPS	USD 41.64	Oper.EPS2024E	USD 45.66	Yield [%]	2.19	3-yr Proj. EPS CAGR[%]	10
Trailing 12-Month P/E	20.13	P/E on Oper.EPS2023E	19.86	Dividend Rate/Share	USD 18.4	SPGMI's Quality Ranking	B+
USD 10K Invested 5 Yrs Ago	46,698.0	Common Shares Outstg.[M]	413.00	Trailing 12-Month Dividend	USD 18.4	Institutional Ownership [%]	82.0

Price Performance



Source: CFRA, S&P Global Market Intelligence
Past performance is not an indication of future performance and should not be relied upon as such.
Analysis prepared by **Angelo Zino, CFA** on Sep 01, 2023 02:36 AM ET, when the stock traded at **USD 880.08**.

Highlights

- ▶ We forecast sales up 7.6% in FY 24 [Oct.] and 4.5% in FY 25 after our forecast for a 7.8% rise in FY 23. Semiconductor Solutions grew 5% in Jul-Q, while Infrastructure Software also grew 5%. Data center build-outs and infrastructure upgrades are driving greater adoption in networking [+20%] for AVGO's switching platforms—Tomahawk, Trident, and Jericho—as well as demand for Ethernet switches for AI [\$200M in FY 22; quadrupling in FY 23]. Enterprise orders are likely to be sustained in the near term, while wireless demand sees near-term seasonal softness. Infrastructure Software sales are predictable with over 90% of bookings recurring and an average contract term length of three years.
- ▶ We see an annual gross margin of about 74% to 75% through FY 25, near the 75.6% margin posted in FY 22. Semiconductor Solutions' gross margin stands at 70% [down 160 bps in Jul-Q], while software stands at 92%. We remain optimistic about operating margin expansion ahead, led by higher volume.
- ▶ We like FCF potential [52% of Jul-Q sales], which we see at \$18B in FY 23 and \$20B in FY 24 [\$16B in FY 22]. Robust FCF is supporting greater shareholder return [dividend hiked by 12% in Dec. 2022 and we see modest buybacks].

Investment Rationale/Risk

- ▶ Our Buy reflects to view AVGO among the biggest potential winners from an AI infrastructure boom, driven by its networking/switcher and ASIC businesses. We also think visibility/demand prospects are better than ever, given the Apple chip supply extension [20% of sales] and AI opportunities looking ahead. If the VMware deal is consummated [software exposure grows to 40%-45% sales post-deal], it is expected to support \$8.5B in targeted EBITDA contribution [combined annual FCF +\$25B]. We see improving networking demand for its switching/routing platforms as well as for its custom ASIC solutions [generative AI seen raising ASIC revenue to >\$3B in FY 23 from about \$2B in FY 22] in cloud while we have high confidence in management execution.
- ▶ Risks to our rating and target price include slowing growth profile, uncertain macro view, elevated debt, and potential loss of Apple business over time [unlikely before 2025].
- ▶ Our 12-month target of \$975 is based on a P/E of 19.9x our CY 25 EPS estimate, above AVGO's three- and five-year historical forward averages of 13.4x and 14.4x to reflect our view of increasing AI opportunities [25% of semiconductor revenue by FY 24, or 18%-20% of total sales].

Analyst's Risk Assessment

LOW MEDIUM HIGH

Our risk assessment reflects our concerns regarding AVGO's highly concentrated customer mix and the company's significant exposure to demand from the volatile consumer electronics market.

Revenue/Earnings Data

Revenue (Million USD)	1Q	2Q	3Q	4Q	Year
2025	E 9,948	E 9,829	E 10,081	E 10,450	E 40,310
2024	E 9,454	E 9,384	E 9,622	E 10,040	E 38,500
2023	8,915	8,733	8,876	E 9,270	E 35,794
2022	7,706	8,103	8,464	8,930	33,203
2021	6,655	6,610	6,778	7,407	27,450
2020	5,858	5,742	5,821	6,467	23,888

Earnings Per Share (USD)

	1Q	2Q	3Q	4Q	Year
2025	E 11.80	E 11.61	E 12.01	E 12.70	E 48.12
2024	E 11.18	E 11.14	E 11.39	E 11.94	E 45.66
2023	10.33	10.32	10.54	E 11.02	E 42.21
2022	8.39	9.07	9.73	10.45	37.64
2021	6.61	6.62	6.96	7.81	28.01
2020	5.25	5.14	5.40	6.35	22.16

Fiscal Year ended Oct 31. EPS Estimates based on CFRA's Operating Earnings; historical earnings are adjusted. In periods where a different currency has been reported, this has been adjusted to match the current quoted currency.

Dividend Data

Amount [USD]	Date Decl.	Ex-Div. Date	Stk. of Record	Payment Date
4.6000	Aug 30	Sep 20	Sep 21	Sep 29 '23
4.6000	May 31	Jun 21	Jun 22	Jun 30 '23
4.6000	Mar 01	Mar 21	Mar 22	Mar 31 '23
4.6000	Dec 06	Dec 19	Dec 20	Dec 30 '22

Dividends have been paid since 2010 . Source: Company reports
Past performance is not an indication of future performance and should not be relied as such.
Forecasts are not a reliable indicator of future performance.
Dividends paid in currencies other than the Trading currency have been accordingly converted for display purposes.



Business Summary Sep 01, 2023

CORPORATE OVERVIEW. Broadcom [AVGO] is a leading designer, developer, and global supplier of a broad range of analog semiconductor devices, with a focus on radio frequency [RF] and optoelectronic applications. AVGO breaks its revenue into two primary segments: Semiconductor Solutions [78% of FY 22 sales] and Infrastructure Software [22% of FY 22 sales].

The company strategically focuses its research and development resources to address niche opportunities in its target markets and leverage its extensive portfolio of the U.S. and other patents, and other intellectual property [IP] to integrate multiple technologies and create system-on-chip [SoC] component and software solutions that target growth opportunities. AVGO develops semiconductor devices with a focus on complex digital and mixed signal complementary metal oxide semiconductor [CMOS] based devices and analog III-V based products.

Semiconductor solutions manage the movement of data in data center, telecom, enterprise, and embedded networking applications. AVGO provides a broad variety of radio frequency semiconductor devices, wireless connectivity solutions, and custom touch controllers for mobile applications. The company also provides semiconductor solutions for enabling the set-top box and broadband access markets, and for enabling secure movement of digital data to and from host machines, such as servers, personal computers, and storage systems, to the underlying storage devices, such as hard disk drives and solid state drives.

AVGO supports the wireless industry with a broad variety of RF semiconductor devices. It was among the first to deliver commercial film bulk acoustic resonator, or FBAR, filters, which offer technological advantages over competing filters for smartphones to function more efficiently in today's congested RF spectrum. FBAR technology historically has maintained a significant market share within code division multiple access, or CDMA. In the storage and Ethernet networking markets, AVGO supplies transceivers that receive and transmit information along optical fibers. In Ethernet networking, AVGO focuses on higher bandwidth applications for data centers as well as enterprise customers. In storage, it focuses on fiber channel-based applications. Broadcom also sells communication solutions for set-top box, cable modem, and carrier access.

Enterprise storage products enable secure movement of digital data to and from host machines such as servers, personal computers, and storage systems to the underlying storage devices such as hard disk drives and solid state drives. Industrial and automotive products include optical isolators, or optocouplers, which provide electrical insulation and signal isolation for signaling systems that are susceptible to electrical noise or interference.

The Infrastructure software segment offers a portfolio of mainframe, enterprise, and storage area networking solutions, which enables customers to utilize the benefits of agility, automation, insights, resiliency, and security in managing business processes and technology investments, and to reduce the cost and complexity of managing business information within a shared data storage environment. AVGO's mainframe software solutions include solutions for the IBM Z mainframe platform, which runs its customers' mission-critical business applications. These software products help customers improve economics by increasing throughput and lowering cost per transaction, increasing business agility through DevOps tooling and processes, increasing reliability and availability of operations through machine intelligence and automation solutions, and protecting enterprise data with security and compliance. The company's enterprise software solutions enable large global organizations to transform into digital businesses by providing a digital infrastructure management platform that delivers speed, agility, and the ability to optimize for risk across multi-cloud hybrid environments.

CORPORATE STRATEGY. AVGO focuses on maintaining an efficient global supply chain and a variable low-cost operating model. Given this, AVGO outsources a majority of its manufacturing operations utilizing third-party foundry and assembly and test capabilities.

IMPACT OF MAJOR DEVELOPMENTS. On May 26, 2022, AVGO announced its intent to acquire VMware for \$61B [half equity/debt] for \$142.50 in cash or 0.252 shares of AVGO shares [88%/12% AVGO/VMW ownership]. Although a deal will leverage AVGO's balance sheet [net debt pro-forma to exceed \$70B], we note the potential for \$8.5B in targeted EBITDA contribution on the VMware side [combined annual FCF likely exceeds \$20B]. We believe a deal makes sense given AVGO's desire to further diversify into the higher margin [VMW's gross margin at 85% vs. AVGO's 75%] and recurring based enterprise software arena [raises software exposure to 45%-50% of sales]. We also think a combination is in AVGO's wheelhouse, as it seeks lower software growth but stable cash generating firms where it could look for significant cost synergies to maximize FCF/EPS.

In November 2019, AVGO acquired Symantec's Enterprise Security business for \$10.7B in cash. In November 2018, AVGO completed its acquisition of CA, Inc. for \$44.50 per share, or \$18.9B, in cash. On November 6, 2017, AVGO announced its intent to acquire QUALCOMM [QCOM] for \$70 per share [\$60 in cash and \$10 AVGO stock]. On November 13, 2017, QCOM rejected AVGO's offer. Ultimately, on March 12, 2018, the Trump administration blocked a combination between the two sides.

FINANCIAL TRENDS. The company generated \$33.2B in revenue during FY 22, growing 21%. Semiconductor Solutions revenue was \$25.8B, up 27%, while Infrastructure Software revenue was \$7.4B, up 4%. Gross margin in FY 22 was 76%, up 110 basis points. FY 22 operating income was \$20.3B, up 28% year-over-year, and represented 61% of net revenue.

AVGO spent \$424M on capital expenditures in FY 22 and free cash flow grew 22% to \$16.3B or 49% of FY 22 revenue. For FY 22, the company spent \$15.5B, consisting of \$7B in the form of cash dividends and \$8.5B in repurchases.

The company had \$12.1B of cash and \$39.3B of gross debt at the end of July.

Corporate information

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Fax N/A	
Website www.broadcom.com	
Officers	
Chairman of the Board H. E. Samuelli	President, CEO, & Executive Director H. E. Tan
Chief Legal & Corporate Affairs Officer M. D. Brazeal	CFO & Chief Accounting Officer K. M. Spears
Board Members	
C. Low	H. E. Tan
D. M. Bryant	H. L. You
E. W. Hartenstein	J. F. Page
G. J. Delly	R. J. Fernandez
H. E. Samuelli	
Domicile Delaware	Auditor PricewaterhouseCoopers LLP
Founded 2018	
Employees 20,000	
Stockholders 1,060	



Broadcom Inc.

Quantitative Evaluations						
Fair Value Rank		1	2	3	4	5
		LOWEST		HIGHEST		
		Based on CFRA's proprietary quantitative model, stocks are ranked from most overvalued (1) to most undervalued (5).				
Fair Value Calculation		USD 691.61	Analysis of the stock's current worth, based on CFRA's proprietary quantitative model suggests that AVGO is overvalued by USD 146.75 or 17.50%			
Volatility		LOW		AVERAGE		HIGH
Technical Evaluation		BEARISH	Since August, 2023, the technical indicators for AVGO have been BEARISH"			
Insider Activity		UNFAVORABLE		NEUTRAL		FAVORABLE

Expanded Ratio Analysis				
	2022	2021	2020	2019
Price/Sales	6.02	8.31	6.16	5.50
Price/EBITDA	10.38	15.49	13.07	12.35
Price/Pretax Income	16.09	33.72	60.25	55.83
P/E Ratio	12.56	18.98	15.78	13.93
Avg. Diluted Shares Outstg. [M]	423.00	429.00	421.00	419.00
Figures based on fiscal year-end price				

Key Growth Rates and Averages				
Past Growth Rate [%]		1 Year	3 Years	5 Years
Net Income		70.65	61.60	46.70
Sales		20.96	13.69	13.49
Ratio Analysis (Annual Avg.)				
Net Margin [%]		34.62	23.85	28.48
% LT Debt to Capitalization		62.30	61.31	55.10
Return on Equity [%]		48.20	29.29	29.83

Company Financials Fiscal year ending Oct 31										
Per Share Data [USD]										
	2022	2021	2020	2019	2018	2017	2016	2015	2014	2013
Tangible Book Value	-67.02	-72.31	-89.32	-73.69	-27.00	-37.29	-52.50	-0.86	-7.75	7.99
Free Cash Flow	39.15	31.76	28.11	23.28	19.72	13.54	7.34	6.53	3.05	1.97
Earnings	26.53	15.00	6.33	6.46	28.48	4.03	-4.57	4.95	1.16	2.19
Earnings (Normalized)	37.64	28.01	22.16	21.29	20.82	16.02	11.45	8.98	4.90	2.89
Dividends	16.90	14.90	13.35	11.20	7.90	4.81	2.52	1.64	1.23	0.88
Payout Ratio [%]	61.17	92.22	187.00	155.00	24.46	97.70	NM	29.91	108.00	35.87
Prices: High	677.76	536.07	387.80	323.20	285.68	259.36	179.42	150.50	90.88	47.21
Prices: Low	415.07	346.66	155.67	213.71	197.46	160.62	114.25	83.50	42.45	30.50
P/E Ratio: High	18.00	19.10	17.50	15.20	13.70	16.20	15.70	16.80	18.50	16.30
P/E Ratio: Low	11.00	12.40	7.00	10.00	9.50	10.00	10.00	9.30	8.70	10.60
Income Statement Analysis [Million USD]										
Revenue	33,203	27,450	23,888	22,597	20,848	17,636	13,240	6,824	4,269	2,520
Operating Income	14,286	8,684	4,361	4,258	5,487	2,689	1,871	1,880	884.00	578.00
Depreciation + Amortization	4,984	6,041	6,905	5,808	4,081	4,737	3,042	962.00	625.00	187.00
Interest Expense	1,737	1,885	1,777	1,444	628.00	454.00	585.00	191.00	110.00	2.00
Pretax Income	12,434	6,765	2,443	2,226	4,545	1,825	-1,107	1,467	342.00	568.00
Effective Tax Rate	7.60	0.40	-21.20	-22.90	-177.90	1.90	-58.00	5.20	9.60	2.80
Net Income	11,495	6,736	2,960	2,724	12,259	1,692	NM	1,364	263.00	552.00
Net Income [Normalized]	7,916	4,269	1,649	1,809	2,774	1,344	932.00	1,078	492.50	371.20
Balance Sheet and Other Financial Data [Million USD]										
Cash	12,416	12,163	7,618	5,055	4,292	11,204	3,097	1,822	1,628	1,008
Current Assets	18,504	16,586	11,895	9,917	9,107	15,823	7,125	3,759	3,835	1,818
Total Assets	73,249	75,570	75,933	67,493	50,124	54,418	49,966	10,515	10,491	3,415
Current Liabilities	7,052	6,281	6,371	6,899	2,338	2,529	3,078	1,119	1,016	414.00
Long Term Debt	39,053	39,401	40,187	30,011	17,493	17,431	13,188	3,826	5,463	N/A
Total Capital	62,687	65,262	65,590	57,768	44,150	40,734	35,518	8,586	8,752	2,886
Capital Expenditures	424.00	443.00	463.00	432.00	635.00	1,069	723.00	593.00	409.00	236.00
Cash from Operations	16,736	13,764	12,061	9,697	8,880	6,551	3,411	2,318	1,175	722.00
Current Ratio	2.62	2.64	1.87	1.44	3.90	6.26	2.31	3.36	3.77	4.39
% Long Term Debt of Capitalization	62.30	60.40	61.30	52.00	39.60	42.80	37.10	44.60	62.40	N/A
% Net Income of Revenue	34.60	24.50	12.40	12.10	58.80	9.60	-13.10	20.00	6.20	21.90
% Return on Assets	12.00	7.16	3.80	4.53	6.56	3.22	3.87	11.19	7.95	11.51
% Return on Equity	48.20	27.60	12.10	10.60	50.70	7.90	-13.20	35.00	10.10	20.80

Source: S&P Global Market Intelligence. Data may be preliminary or restated; before results of discontinued operations/special items. Per share data adjusted for stock dividends; EPS diluted. E-Estimated. NA-Not Available. NM-Not Meaningful. NR-Not Ranked. UR-Under Review.

Broadcom Inc.

Sub-Industry Outlook

We have a positive fundamental outlook for the semiconductors sub-industry over the next 12 months. We expect semiconductor sales to rebound about 10% in 2024 after our forecast for a 12% decline in 2023, with a cyclical recovery to start in the second half of the year [Y/Y revenue growth returns in Q4 2023]. We are starting to see a return to sequential growth across the more cyclical areas of the semiconductor market (PCs, smartphones, and gaming), driven in part by seasonality as well as a more normalized inventory channel after sharply undershipping end-demand for the last several quarters. Whether or not recent strength can be sustained will be dependent on macro factors and the health of the consumer, but we think the industry is in a very healthy situation.

Currently, the biggest question mark is within the industrial/automotive markets as those areas held up well through 1H 2023, but channel inventory is now being rightsized due to end-market softness, which may continue until at least Q1 2024. Although inventories are still above optimal levels, companies are making efforts (e.g., production cuts) to quickly get back to normalized levels. In the long term, we expect content growth in automobiles to sharply outpace semiconductor industry growth, driven by several megatrends, such as autonomy, electrification, and connected infotainment. Analog semiconductors are particularly exposed to these trends.

Although data center spending has been relatively muted in 2023 (we estimate ~2%), we see a 10%-15% rebound in 2024, with investments continuing to be focused on upgrading the network to meet more artificial intelligence (AI) driven workloads. As a result, we expect accelerated computing technologies (GPUs, FPGAs, and ASICs) to gain traction, in our view, as they help overcome the limitations of CPUs. We believe that the recent spend shift towards accelerators will be permanent and will grow to an annual run rate of over \$130 billion by 2027 versus our estimated \$45 billion-\$50 billion run rate in 2023. Chipmakers tied to the data center are exposed to the best financially positioned

customers on the planet, and despite quarterly lumpiness, the long-term prospects related to themes like AI and the metaverse remain extremely encouraging and offer significant growth potential ahead.

We believe the semiconductor industry will witness a super cycle over the next decade as content growth per device across a host of markets accelerates, driven by the emergence of greater AI capabilities. Given this, we estimate the semiconductors industry will grow at a 10-year annualized pace of 6%-7% (using 2021 expected revenue as the base year), or at least 2x global GDP.

While companies that outsource manufacturing would typically have more stability in gross margins (a trade-off for capped upside), others with their own manufacturing see more variability. Industry margins are a function of manufacturing utilization and inventory supply-demand imbalances. We forecast some companies, especially in analog, to maintain higher inventory levels. Healthy free cash flow generation and high cash levels will likely result in greater shareholder return via share repurchases. Regulatory and geopolitical uncertainties remain notable risks.

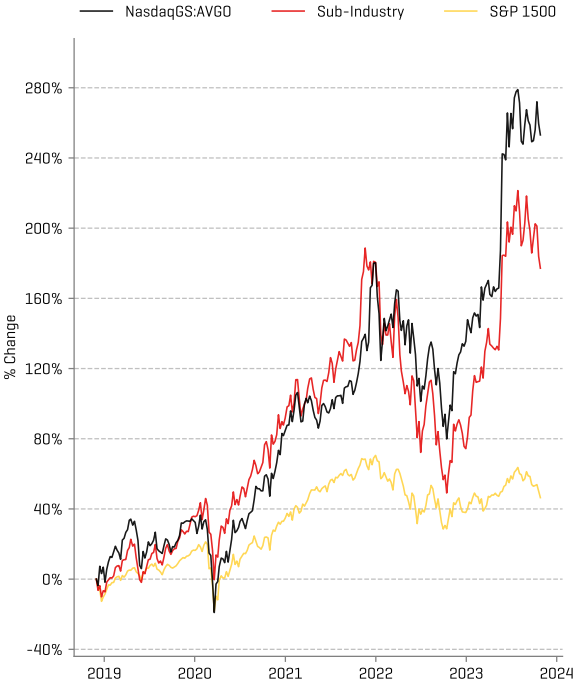
YTD through September 22, the S&P semiconductors sub-industry index rose 64.0% vs. an 11.6% increase for the S&P 1500 index. For full-year 2022, the S&P semiconductors sub-industry index fell 37.9% vs. a 19.1% decline for the S&P 1500 index.

/ Angelo Zino, CFA

Industry Performance

GICS Sector: Information Technology
Sub-Industry: Semiconductors

Based on S&P 1500 Indexes
Five-Year market price performance through Oct 28, 2023



NOTE: A sector chart appears when the sub-industry does not have sufficient historical index data.
All Sector & Sub-Industry information is based on the Global Industry Classification Standard (GICS).
Past performance is not an indication of future performance and should not be relied upon as such.
Source: CFRA, S&P Global Market Intelligence

Sub-Industry: Semiconductors Peer Group*: Semiconductors

Peer Group	Stock Symbol	Exchange	Currency	Recent Stock Price	Stk. Mkt. Cap. (M)	30-Day Price Chg. (%)	1-Year Price Chg. (%)	P/E Ratio	Fair Value Calc.	Yield (%)	Return on Equity (%)	LTD to Cap (%)
Broadcom Inc.	AVGO	NasdaqGS	USD	826.75	341,229.0	1.3	79.3	20.0	691.61	2.2	64.8	62.2
Advanced Micro Devices, Inc.	AMD	NasdaqGS	USD	93.67	151,340.0	-2.4	56.8	37.0	N/A	N/A	-0.0	3.0
Analog Devices, Inc.	ADI	NasdaqGS	USD	160.86	80,159.0	-6.7	13.8	15.0	154.34	2.1	10.3	15.0
Arm Holdings plc	ARM	NasdaqGS	USD	50.17	51,478.0	-6.3	N/A	129.0	N/A	N/A	N/A	N/A
Intel Corporation	INTC	NasdaqGS	USD	32.52	136,194.0	-3.9	19.5	42.0	N/A	1.5	-0.9	30.2
Marvell Technology, Inc.	MRVL	NasdaqGS	USD	46.63	40,232.0	-9.8	20.3	28.0	N/A	0.5	-2.4	15.8
Micron Technology, Inc.	MU	NasdaqGS	USD	64.53	70,856.0	-5.0	16.4	NM	N/A	0.7	-12.4	20.5
NXP Semiconductors N.V.	NXPI	NasdaqGS	USD	179.56	46,291.0	-7.9	21.9	13.0	N/A	2.3	35.8	51.8
QUALCOMM Incorporated	QCOM	NasdaqGS	USD	105.62	117,872.0	-3.7	-10.6	11.0	132.90	3.0	47.0	40.2
Taiwan Semiconductor Manufacturing Company Limited	TSM	NYSE	USD	87.45	424,160.0	3.8	43.7	16.0	N/A	1.7	29.2	21.7
Texas Instruments Incorporated	TXN	NasdaqGS	USD	144.01	130,790.0	-8.8	-8.8	19.0	138.71	3.6	45.6	39.2

*For Peer Groups with more than 10 companies or stocks, selection of issues is based on market capitalization.
NA-Not Available; NM-Not Meaningful.
Note: Peers are selected based on Global Industry Classification Standards and market capitalization. The peer group list includes companies with similar characteristics, but may not include all the companies within the same industry and/or that engage in the same line of business.

Broadcom Inc.

Analyst Research Notes and other Company News

September 21, 2023

09:24 AM ET... CFRA Maintains Buy Opinion on Shares of Broadcom Inc. [AVGO 830.57****]:

According to an unconfirmed report from The Information, Alphabet [GOOGL 134 ****] has discussed dropping AVGO as a supplier of its tensor processing units as soon as 2027 given pricing concerns between the two sides. If true, we note the move would be a major blow to AVGO's AI initiatives, as GOOGL represents the only major customer within its custom silicon business, estimated to rise more than 50% in FY 2023 to +\$3B [estimated at roughly 8% of its current revenue]. Although the uncertainty creates an overhang on shares, we think the business will continue to grow over the next several years and that AVGO will likely have the opportunity to expand its custom silicon business to other hyperscalers if it so chooses. We note that AVGO's AI chip exposure [networking + custom silicon] is accounting for all its growth on the semiconductor side amid tough macro conditions. Separately, a completion of the VMware deal would further diversify AVGO from its large chip customers, like GOOGL and Apple [AAPL 175 ****]. / Angelo Zino, CFA

September 01, 2023

01:22 AM ET... CFRA Maintains Buy Opinion on Shares of Broadcom Inc. [AVGO 880.08****]:

We keep our 12-month target price of \$975 on a P/E of about 19.9x our CY 25 EPS view, above historical given AI prospects and EPS upside from the VMware deal [closing by 10/31]. We keep our FY 23 [Oct.] EPS estimate at \$42.21 and FY 24 at \$45.66. We start FY 25 at \$48.12. AVGO posts Jul-Q EPS of \$10.54 vs. \$9.73, beating the \$10.43 consensus. Sales grew 4.9%, near expectations, on +5% growth in Semiconductor Solutions and similar pace within Infrastructure Software. We attribute all the growth on the semi side to greater generative AI-related cloud spend while enterprise and telco growth moderated. We think networking remains the key contributor to growth [+20%; 40% of semi sales], as AVGO is seeing healthy momentum on the custom silicon side [v5 TPUs ramp from Alphabet] as well as for its Ethernet switches/routers. We like AVGO's best-in-class FCF [52% of sales], improving inventory position [DIOs down to 80 days from 86 in Apr-Q], and growing AI opportunities [seen +50% seq. in Oct-Q or more than 2x Y/Y]. / Angelo Zino, CFA

June 16, 2023

12:47 AM ET... CFRA Maintains Buy Opinion on Shares of Broadcom Inc. [AVGO 883.43****]:

We lift our 12-month target to \$975 from \$850 on a P/E of about 21x our CY 24 EPS view, above historical given AI prospects. We lift our FY 23 [Oct.] EPS estimate to \$42.21 from \$41.84 and FY 24 to \$45.66 from \$44.51. Our more aggressive target price reflects our belief that the accelerator market will grow significantly faster than we previously forecasted [+ \$130B by 2027 from + \$60B in our March forecast; about \$30B-\$35B seen in 2023], demonstrating how much the landscape has changed in the last 75 days. This offers significant upside to both AVGO's compute offload and networking businesses. On the compute offload side [\$3B in '23 from \$2B in '22], where Alphabet is its only major customer, we do see potential to grow/expand at this existing customer and to other hyperscalers. On networking [quadrupling to \$800M revenue in '23], we think this will be the primary AI growth driver and ultimately eclipses compute offload revenue. We estimate AI revenue growing to 20% of total sales in CY 24 [low-teens in CY 23]. / Angelo Zino, CFA

June 02, 2023

12:27 AM ET... CFRA Maintains Buy Opinion on Shares of Broadcom Inc. [AVGO 789.95****]:

We lift our 12-month target to \$850 from \$750 on a P/E of about 18.9x our CY 24 EPS view, above historical to reflect our view of AI opportunities. We lift our FY 23 [Oct.] EPS estimate to \$41.84 from \$41.35 and FY 24 to \$44.51 from \$43.58. AVGO posts Apr-Q EPS of \$10.32 vs. \$9.07, beating the \$10.14 consensus estimate. Sales rose 8%, near our view, with growth in Semiconductor Solutions [+9%] and Infrastructure Software [+3%]. We view AVGO as among the biggest potential winners from an AI infrastructure boom, driven by its networking/switcher business and ASIC Solutions. AI revenue is seen growing to about 25% of semiconductor revenue by FY 24 [18%-20% of total sales] vs. 15% currently [assuming no VMware]. We think visibility/demand prospects across AVGO's semiconductor business are better than its ever been following the Apple chip supply extension [wireless orders to ramp in the next 3-6 months] and given AI opportunities ahead. Despite net debt of \$27.7B, we note FCF [50% of revenue] is the best in semis. / Angelo Zino, CFA

May 23, 2023

02:48 PM ET... CFRA Maintains Buy Opinion on Shares of Broadcom Inc. [AVGO 692.68****]:

We raise our 12-month target to \$750 from \$680 on a higher revised P/E of about 17x our CY 2024 EPS estimate, above historical levels to reflect our view of AI opportunities, but below peers given its more leveraged balance sheet. AVGO announces the extension of its chip supply agreement with Apple [AAPL 172 ****], and while terms weren't disclosed, it is for multiple years. We positively view the extension given AAPL's recent endeavors to internally source more chips and rumors earlier this year that certain AVGO chips could be displaced [in 2025]. We also note that AAPL is AVGO's largest customer, about 20% of sales over the last year. We like AVGO's story, largely given opportunities that are evolving in the cloud, specifically as it relates to its AI initiatives – both on the network connectivity side and as a co-developer of application specific chips [e.g., Alphabet's TPUs]. Despite the uncertainty surrounding the pending VMware deal, we think shares have upside whether a deal is done or not. / Angelo Zino, CFA

April 19, 2023

02:10 AM ET... CFRA Adds Shares of Broadcom Inc. to its U.S. Women's Participation Portfolio [AVGO 632.32****]:

Our Buy recommendation on the shares of this well-diversified designer, developer, and global supplier of analog semiconductor devices reflects our growing confidence in opportunities AVGO has in the cloud, which is more than offsetting our concerns over the impact of a slowing wireless business and the uncertainty surrounding AVGO's pending VMware acquisition. Our 12-month target price of \$680 values AVGO shares at 15.5x our CY 24 EPS view of \$43.78, a discount to peers, reflecting AVGO's above-peer financial leverage. Broadcom replaces L3Harris Technologies Inc. [LHX] in the U.S. Women's Participation Portfolio, following the downgrade of LHX shares to Hold. / Catherine Seifert

March 03, 2023

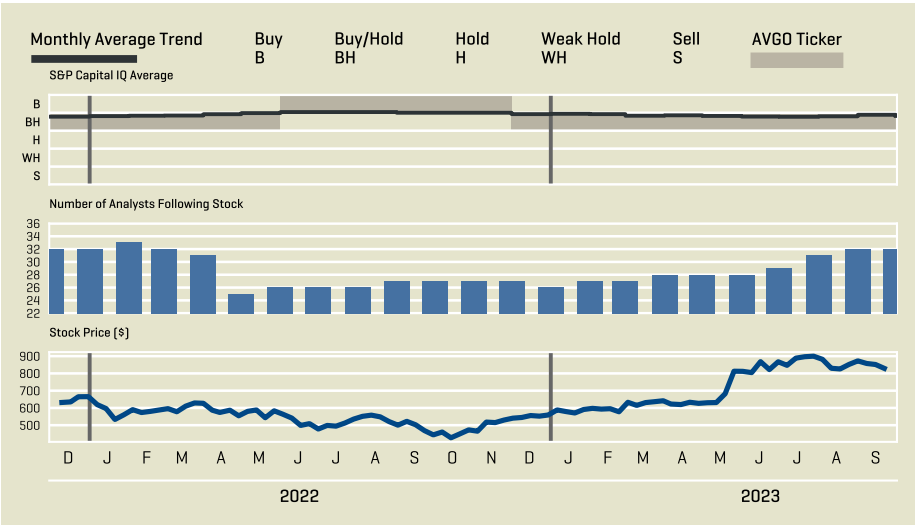
10:09 AM ET... CFRA Upgrades Opinion on Shares of Broadcom Inc. to Buy from Hold [AVGO 598.65****]:

We up our 12-month target to \$680 from \$600, on a P/E of 15.5x our CY 24 EPS view of \$43.78, below peers to reflect AVGO's leveraged financial position. We raise our FY 23 [Oct.] EPS estimate to \$41.35 from \$41.05 and FY 24 to \$43.58 from \$42.59. MRVL posts Jan-Q EPS of \$10.33 vs. \$8.39, beating the \$10.16 estimate. Sales rose 16%, as 21% in Semiconductor Solutions was partly offset by a 1% decline in Infrastructure Software. Gross margin compressed to 73.8% from 75.5%, on less favorable mix. We believe AVGO has good visibility, with lead times around 50 weeks and inventories at healthy levels. We are impressed with networking growth [+20%], on greater Tomahawk deployment and demand for Ethernet switches for AI [\$200M in FY 22 to \$800M in FY 23]. Generative AI is also helping to support demand for AVGO's ASIC business [\$3B in FY 23 from \$2B in FY 22]. Our confidence about growing opportunities in the cloud more than offsets our concerns about a slowing wireless business and uncertainty around the VMware deal. / Angelo Zino, CFA

Note: Research notes reflect CFRA's published opinions and analysis on the stock at the time the note was published. The note reflects the views of the equity analyst as of the date and time indicated in the note, and may not reflect CFRA's current view on the company.

Broadcom Inc.

Analysts Recommendations



	No. of Recommendations	% of Total	1 Mo.Prior	3 Mos.Prior
Buy	19	59	20	17
Buy/Hold	4	13	4	4
Hold	7	22	6	7
Weak hold	1	3	1	1
Sell	0	0	0	0
No Opinion	1	3	1	0
Total	32	100	32	29

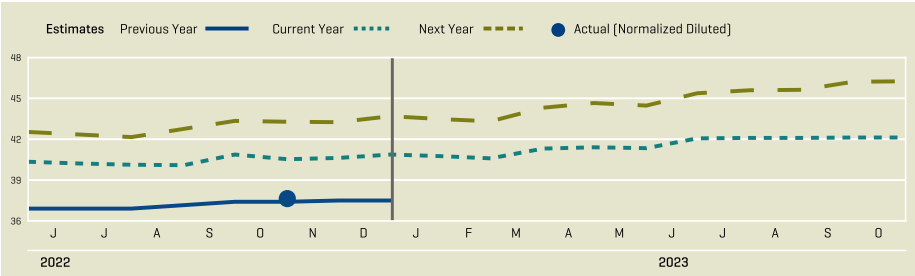
Wall Street Consensus Opinion

Buy/Hold

Wall Street Consensus vs. Performance

For fiscal year 2023, analysts estimate that AVGO will earn USD 42.12. For fiscal year 2024, analysts estimate that AVGO's earnings per share will grow by 9.78% to USD 46.24.

Wall Street Consensus Estimates



Fiscal Year	Avg Est.	High Est.	Low Est.	# of Est.	Est. P/E
2024	46.24	49.63	43.40	26	17.88
2023	42.12	42.95	41.53	27	19.63
2024 vs. 2023	▲ 10%	▲ 16%	▲ 4%	▼ -4%	▼ -9%
Q4'24	12.09	13.61	11.24	19	68.40
Q4'23	10.95	11.08	10.71	25	75.49
Q4'24 vs. Q4'23	▲ 10%	▲ 23%	▲ 5%	▼ -24%	▼ -9%

Forecasts are not reliable indicator of future performance.
Note: A company's earnings outlook plays a major part in any investment decision. S&P Global Market Intelligence organizes the earnings estimates of over 2,300 Wall Street analysts, and provides their consensus of earnings over the next two years, as well as how those earnings estimates have changed over time. Note that the information provided in relation to consensus estimates is not intended to predict actual results and should not be taken as a reliable indicator of future performance.
Note: For all tables, graphs and charts in this report that do not cite any reference or source, the source is S&P Global Market Intelligence.

Broadcom Inc.

Glossary

STARS

Since January 1, 1987, CFRA Equity and Fund Research Services, and its predecessor S&P Capital IQ Equity Research has ranked a universe of U.S. common stocks, ADRs [American Depositary Receipts], and ADSs [American Depositary Shares] based on a given equity's potential for future performance. Similarly, we have ranked Asian and European equities since June 30, 2002. Under proprietary STARS [Stock Appreciation Ranking System], equity analysts rank equities according to their individual forecast of an equity's future total return potential versus the expected total return of a relevant benchmark [e.g., a regional index (MSCI AC Asia Pacific Index, MSCI AC Europe Index or S&P 500® Index)], based on a 12-month time horizon. STARS was designed to help investors looking to put their investment decisions in perspective. Data used to assist in determining the STARS ranking may be the result of the analyst's own models as well as internal proprietary models resulting from dynamic data inputs.

S&P Global Market Intelligence's Quality Ranking

[also known as **S&P Capital IQ Earnings & Dividend Rankings**] - Growth and S&P Capital IQ Earnings & Dividend Rankings stability of earnings and dividends are deemed key elements in establishing S&P Global Market Intelligence's earnings and dividend rankings for common stocks, which are designed to capsize the nature of this record in a single symbol. It should be noted, however, that the process also takes into consideration certain adjustments and modifications deemed desirable in establishing such rankings. The final score for each stock is measured against a scoring matrix determined by analysis of the scores of a large and representative sample of stocks. The range of scores in the array of this sample has been aligned with the following ladder of rankings:

A+ Highest	B Below Average
A High	B- Lower
A Above	C Lowest
B+ Average	D In Reorganization
NC Not Ranked	

EPS Estimates

CFRA's earnings per share (EPS) estimates reflect analyst projections of future EPS from continuing operations, and generally exclude various items that are viewed as special, non-recurring, or extraordinary. Also, EPS estimates reflect either forecasts of equity analysts; or, the consensus [average] EPS estimate, which are independently compiled by S&P Global Market Intelligence, a data provider to CFRA. Among the items typically excluded from EPS estimates are asset sale gains; impairment, restructuring or merger-related charges; legal and insurance settlements; in process research and development expenses; gains or losses on the extinguishment of debt; the cumulative effect of accounting changes; and earnings related to operations that have been classified by the company as discontinued. The inclusion of some items, such as stock option expense and recurring types of other charges, may vary, and depend on such factors as industry practice, analyst judgment, and the extent to which some types of data is disclosed by companies.

12-Month Target Price

The equity analyst's projection of the market price a given security will command 12 months hence, based on a combination of intrinsic, relative, and private market valuation metrics, including Fair Value.

Abbreviations Used in Equity Research Reports

- CAGR - Compound Annual Growth Rate
- CAPEX - Capital Expenditures
- CY - Calendar Year
- DCF - Discounted Cash Flow
- DDM - Dividend Discount Model
- EBIT - Earnings Before Interest and Taxes
- EBITDA - Earnings Before Interest, Taxes, Depreciation & Amortization
- EPS - Earnings Per Share
- EV - Enterprise Value
- FCF - Free Cash Flow
- FFO - Funds From Operations
- FY - Fiscal Year
- P/E - Price/Earnings
- P/NAV - Price to Net Asset Value
- PEG Ratio - P/E-to-Growth Ratio
- PV - Present Value
- R&D - Research & Development
- ROCE - Return on Capital Employed
- ROE Return on Equity
- ROI - Return on Investment
- ROIC - Return on Invested Capital
- ROA - Return on Assets
- SG&A - Selling, General & Administrative Expenses
- SOTP - Sum-of-The-Parts
- WACC - Weighted Average Cost of Capital

Dividends on American Depositary Receipts (ADRs) and American Depositary Shares (ADSs) are net of taxes (paid in the country of origin).

Qualitative Risk Assessment

Reflects an equity analyst's view of a given company's operational risk, or the risk of a firm's ability to continue as an ongoing concern. The Qualitative Risk Assessment is a relative ranking to the U.S. STARS universe, and should be reflective of risk factors related to a company's operations, as opposed to risk and volatility measures associated with share prices. For an ETF this reflects on a capitalization-weighted basis, the average qualitative risk assessment assigned to holdings of the fund.

STARS Ranking system and definition:

★★★★★ 5-STARS (Strong Buy):

Total return is expected to outperform the total return of a relevant benchmark, by a notable margin over the coming 12 months, with shares rising in price on an absolute basis.

★★★★★ 4-STARS (Buy):

Total return is expected to outperform the total return of a relevant benchmark over the coming 12 months.

★★★★★ 3-STARS (Hold):

Total return is expected to closely approximate the total return of a relevant benchmark over the coming 12 months.

★★★★★ 2-STARS (Sell):

Total return is expected to underperform the total return of a relevant benchmark over the coming 12 months.

★★★★★ 1-STAR (Strong Sell):

Total return is expected to underperform the total return of a relevant benchmark by a notable margin over the coming 12 months, with shares falling in price on an absolute basis.

Relevant benchmarks:

In North America, the relevant benchmark is the S&P 500 Index, in Europe and in Asia, the relevant benchmarks are the MSCI AC Europe Index and the MSCI AC Asia Pacific Index, respectively.



Broadcom Inc.

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Stocks are ranked in accordance with the following ranking methodologies:

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Quantitative Stock Reports:

Quantitative rankings are determined by ranking a universe of common stocks based on 5 measures or model categories: Valuation, Quality, Growth, Street Sentiment, and Price Momentum. In the U.S., a sixth sub-category for Financial Health will also be displayed. Percentile scores are used to compare each company to all other companies in the same universe for each model category. The five [six] model category scores are then weighted and rolled up into a single percentile ranking for that company. For reports containing quantitative rankings refer to the Glossary section seof the report for detailed methodology and the definition of Quantitative rankings.

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STARS Stock Reports:

Global STARS Distribution as of September 30, 2023

Ranking	North America	Europe	Asia	Global
Buy	36.8%	34.5%	41.6%	37.4%
Hold	53.8%	51.2%	50.3%	52.6%
Sell	9.4%	14.3%	8.1%	10.1%
Total	100.0%	100.0%	100.0%	100.0%

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Broadcom Inc.

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